

Expert advice

► "FPGA and ASIC industries together constitute VLSI design. For those looking at a career in this industry, keeping abreast about the latest developments in the field of SOC's, communication processors and FPGA architectures will be useful." —**Rajesh Chakkingal, lead architect, Mistral Solutions**

► "We have seen several times that candidates do not have a solid grip on basic concepts. Probably, they might have studied only to clear their examination. My advice to new graduates would be to pay good attention to the concepts. Other thing would be to try to do a graduation/post-graduation project related to FPGAs. Pick a good project and do it completely including coding, synthesis and verification. With this, you will be able to learn about the FPGA and its related work, and it could lead to a good FPGA job." —**Yogesh Soni, senior technical director, Mattozetta Technologies**

► "It is necessary that an FPGA engineer plans his/her career for all-round growth. It is important that in the first 2 to 3 years of his/her career, the FPGA engineer focusses on his/her horizontal growth, that is, to get hands on with different aspects of the development cycle. Even exposure in board schematic design, bill of materials (BOMs) creation, board-level debugging using CRO and logic analysers will help in the long term to understand the system-level requirements and work on complete FPGA development cycle." —**Paramita Kapat, head human resource, elfnchips**

► "FPGA technology is getting extremely popular and important. Focussing on the areas of digital design skills, VHDL/Verilog coding and strong analytical skills will definitely increase employability chances and help increase growth in this field." —**Rajesh Choudhary, HR head, Xilinx India**

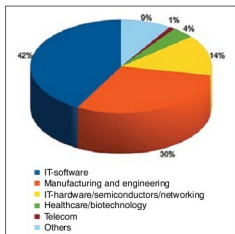


Fig. 1: Top five industries hiring for work in FPGA domain (Data courtesy: TimesJobs.com)

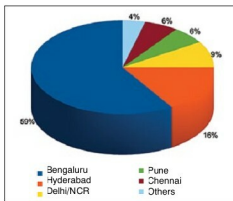


Fig. 2: Top five locations with FPGA jobs (Data courtesy: TimesJobs.com)

hary says, "Today, many companies are using FPGAs for prototyping many ASIC/ASSP applications, hence the opportunities are huge in this field."

As an FPGA engineer, one is expected to handle design, verification and validation tasks. As a fresher, it is necessary to start with training in HDL and FPGA tools.

"At entry level, it is preferred that the engineer works on verification and validation of the design. Identifying and evaluating various scenarios to test, creating test environment and verifying and debugging the design can offer them a good overview of the development cycle," notes Kapat.

She further adds, "They can further

mature as a designer, by owning and designing less complex modules to begin with."

There are many other roles available for FPGA engineers, such as IP design for FPGA, IP porting onto FPGA, FPGA validation, system prototyping using FPGAs and silicon validation using FPGA-based boards.

Demand areas

As per TimesJobs.com's statistics, 42 per cent of FPGA-related jobs are offered by the IT industry, which sort of justifies Soni's statement—"If you hate to do coding, FPGA job is not for you." Thirty per cent of the jobs are available in the manufacturing and engineering sector, 14 per cent in the semiconductor/IT hardware/networking sector

and about 4 per cent of the jobs are available in the healthcare sector.

Geographically speaking, majority of the jobs, which in numbers turns out to be 59 per cent, are in Bengaluru followed by Hyderabad having 16 per cent of the jobs. Delhi currently has about 9 per cent of the jobs and Pune and Chennai have 6 per cent each.

Skills expected

A fresher graduate needs to be strong in basic fundamental areas such as digital electronics, digital logic design (RTL) and microprocessors. A good grasp over languages such as VHDL, Verilog and C is an added advantage. Some firms also expect potential employees to be knowledgeable about digital signal processing (DSP) and communication systems while recruiting them for work in the FPGA domain.

HDL coding skill is a must in any FPGA job. Highlighting the importance of coding in the FPGA domain, Soni says, "If you hate to do coding, an FPGA job is not for you. Besides coding, a good understanding of digital design concepts, logic gates, logic design and verification would help any FPGA engineer to do a good job, to shine and advance in his/her career."

Good problem-solving skills enable an FPGA engineer to analyse different solutions to a given problem and zero in on designing the solution. A good debugging skill makes one quickly find the design bugs and allows a designer to fix it. He adds, "Though this skill comes only with experience, a strong analytical ability helps an engineer to build expertise quickly."

Talking about experienced graduates, Kapat says, "For experienced graduates already working in this field, we strongly rely on their digital design fundamentals. If the experience level is of around five years, we expect the engineer to have worked with complete FPGA development cycle. It is expected that the engineer is well acquainted with FPGA development tools from any one of the FPGA vendors—preferably Xilinx/Altera,